

Technical Document #433: Retrieval Augmented Generation - Comprehensive Overview

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Context window optimization selects the most relevant information to fit within the LLM's context limit. It maximizes information density.

Evaluation metrics for RAG include faithfulness, answer relevance, and context recall. They measure the quality of the retrieval-generation pipeline.

Semantic search finds documents based on meaning rather than exact keywords. It uses embedding similarity to match queries with relevant content.

Vector databases store dense embeddings for semantic search. Pinecone, Weaviate, and ChromaDB are popular vector database solutions.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Key Takeaways:

- Hallucination detection identifies when LLMs generate factually incorrect information.
- Retrieval-Augmented Generation (RAG) combines information retrieval with text generation.
- Evaluation metrics for RAG include faithfulness, answer relevance, and context recall.